Quantum computational complexities in Hamiltonian and topological problems

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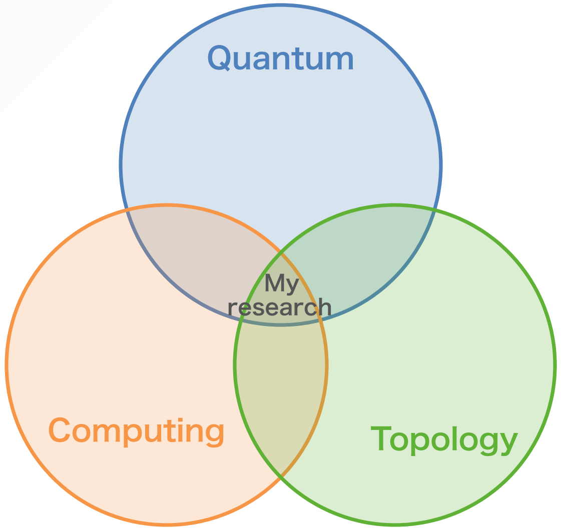


Research interests

- Quantum algorithms & complexity
- Hamiltonian complexity
- Quantum topological data analysis (QTDA)

Lecture overview

Part 1: Quantum Computation Basics

1. The quantum computing model
2. Complexity classes: P, BQP, QMA, ...

Part 2: Hamiltonian Complexity
3. The Local Hamiltonian problem
4. The complexity landscape of Hamiltonians

Part 3: Quantum Topological Data Analysis

1. Classical TDA and the LGZ algorithm
2. The complexity landscape of QTDA

Goal: Understand basic ideas in quantum computational complexity. Understand complexity classes through complete problems in Local Hamiltonian problems, and give an overview of recent advances.

Part 1

Quantum Computation Basics

Quantum circuits

Qubits

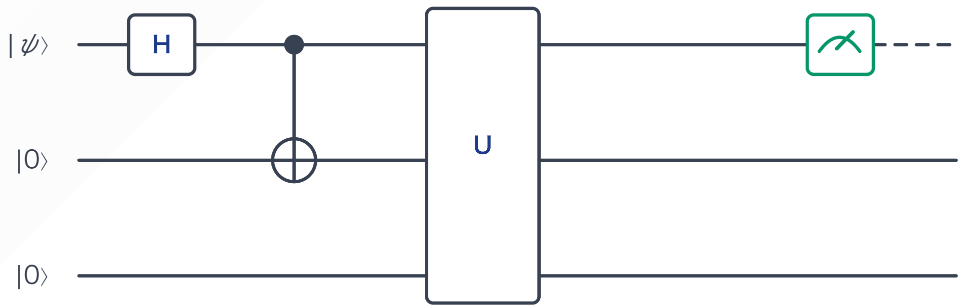
$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1$$

Gates — unitary operations

- Single-qubit: H, X, T, ...
- Two-qubit: CNOT, CZ, ...
- ...

Measurement

$$p(x) = |\langle x|\phi\rangle|^2$$



Key algorithmic primitives

Today's focus is *hardness*, rather than easiness. But we can keep in mind these algorithms:

Primitive
QPE (Quantum Phase Estimation)
Amplitude Amplification
QSVT

Quantum Phase Estimation (QPE)

Goal: given unitary U and eigenstate $U|u\rangle = e^{2\pi i\varphi}|u\rangle$, estimate φ.

$$|0\rangle^{\otimes t} \otimes |u\rangle \xrightarrow{\text{QPE}} |\tilde{\varphi}\rangle \otimes |u\rangle$$

- Uses $O(\log(1/\varepsilon))$ ancilla qubits for precision ε
- Key subroutine: **QFT** (Quantum Fourier Transform)
- Cost: $O(1/\varepsilon)$ applications of controlled-U

Why it matters for us: estimating eigenvalues of H = estimating ground-state energy.

This is relevant to the k-Local Hamiltonian problem (Part 2).

Amplitude Amplification

Generalizes Grover's search: given $|\psi\rangle = \sin\theta\,|\text{good}\rangle + \cos\theta\,|\text{bad}\rangle$,

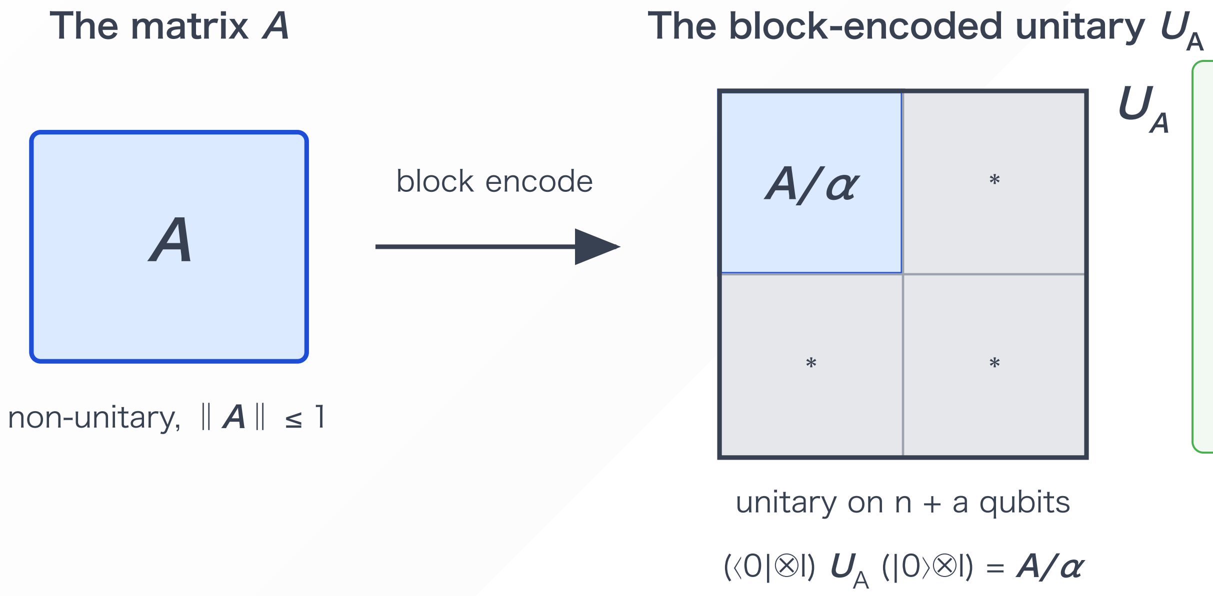
$$O\left(\frac{1}{\sin\theta}\right) \text{ steps } \longrightarrow |\text{good}\rangle \text{ with high probability}$$

- **Quadratic speedup** over classical: $O(1/\sin^2\theta) \to O(1/\sin\theta)$
- Grover: $\sin\theta = 1/\sqrt{N} \to O(\sqrt{N})$ queries
- Useful for e.g. preparing ground states from trial states

QSVT: Block Encoding

Key idea: embed a non-unitary matrix A as the top-left block of a unitary U_A

$$(\langle 0|^{\otimes a} \otimes I)\, U_A \,(|0\rangle^{\otimes a} \otimes I) = \frac{A}{\alpha}$$



- a ancilla qubits
- α: normalization factor
- **Example**: sparse H block-encoded via oracle queries

QSVT: Quantum Singular Value Transformation

Given block encoding U_A and polynomial p with $|p(x)| \leq 1$ on $[-1, 1]$:

$$\text{construct a block encoding of } p(A/\alpha) \text{ using } \deg(p) \text{ calls to } U_A$$

via alternating phase rotations (**Quantum Signal Processing**).

The "grand unification" (Gilyén et al. 2019; Martyn et al. 2021):

Algorithm	Polynomial $p(x)$
Hamiltonian simulation e^{iHt}	Jacobi–Anger approx. of e^{itx}
Amplitude amplification	Chebyshev-type
HHL (linear systems)	$x \mapsto 1/x$

Construction of block-encoding of projector onto ground state

Part 1b

Complexity Classes

Classical hardness: P and NP

P — solvable in polynomial time

- Sorting, shortest path, ⋯

NP — *verifiable* in polynomial time

- SAT, graph coloring, ⋯

NP-complete — the hardest problems in NP

- Cook–Levin theorem: SAT is NP-complete
- If any NP-complete problem is in P, then **P = NP**

BQP — what quantum computers can do

BQP (Bounded-error Quantum Polynomial time):
problems solvable by a poly-time quantum algorithm with error $\leq 1/3$.

Known to be in BQP:

- **Shor's algorithm**: integer factoring

PSPACE (Polynomial Space): problems solvable in polynomial *space* (possibly exponential time).

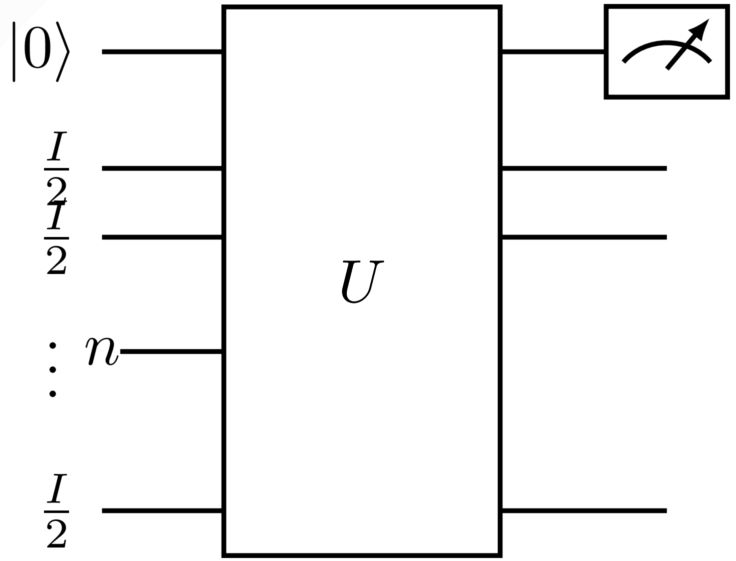
$$\mathbf{P} \subseteq \mathbf{BQP} \subseteq \mathbf{PSPACE}$$

Is **BQP** $\supsetneq$ **BPP**? Believed yes, but not proven. (This implies **PSPACE** $\supsetneq$ **P**.)

DQC1 — one clean qubit

DQC1: problems solvable by a circuit on $n+1$ qubits where the first qubit is $|0\rangle$ and the remaining n are maximally mixed ($I/2^n$), measuring the first qubit at the end.

- Incomparable with BPP (believed)



QMA — quantum analogue of NP

QMA (Quantum Merlin-Arthur):
YES instance has a **quantum witness** $|\psi\rangle$ that a poly-time quantum verifier accepts with prob. $\geq 2/3$;
NO instances rejected with prob. $\geq 2/3$.

$$\mathbf{NP} \subseteq \mathbf{QMA}, \qquad \mathbf{BQP} \subseteq \mathbf{QMA}$$

QMA-complete problem: the k-Local Hamiltonian problem

$$\text{"Is } \lambda_{\min}(H) \leq a \text{ or } \geq b\text{?"}$$

→ the topic of **Part 2**.

Variants of QMA

<u>**QMA_1** (perfect completeness)</u>

YES instances accepted with prob. **1** (completeness = 1); soundness $\leq 1/3$ as usual.

- The power of **QMA_1 depends on the gate set**: different universal gate sets may give different classes

<u>**QCMA** (classical witness)</u>

QCMA: like QMA but the witness is restricted to a **classical bit string**.

$$\mathbf{MA} \subseteq \mathbf{QCMA} \subseteq \mathbf{QMA}$$

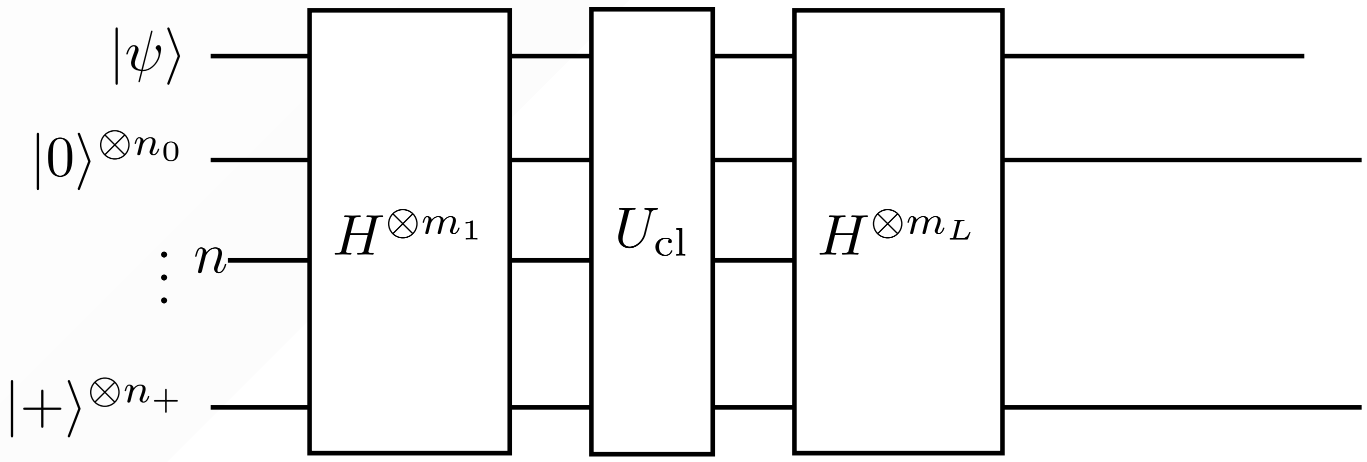
Class	Witness	Verifier
MA	classical	classical (BPP)
QCMA	**classical**	**quantum**
QMA	quantum	quantum

StoqMA — between MA and QMA

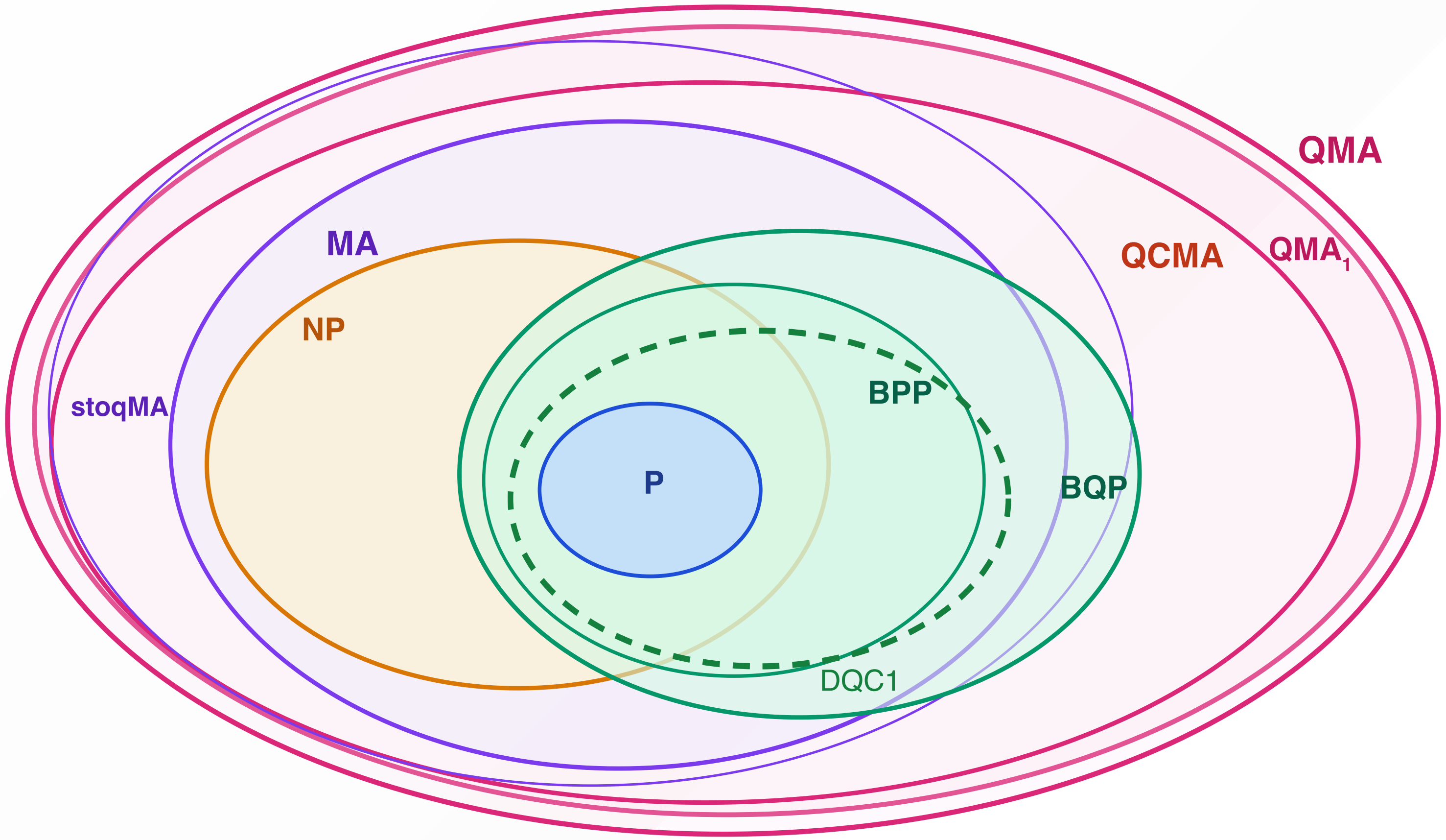
StoqMA: QMA where the verifier uses only a *stoquastic circuit* (X, CNOT, Toffoli + Hadamard layers), measuring in the $\{|+\rangle, |-\rangle\}$ basis.

- Optimal witness has **non-negative amplitudes** → no sign problem → classically interpretable

$$\mathbf{MA} \subseteq \mathbf{StoqMA} \subseteq \mathbf{QMA}$$



The complexity landscape



Key relations:

$$\mathbf{P} \subseteq \mathbf{BQP} \subseteq \mathbf{QMA}$$

$$\mathbf{P} \subseteq \mathbf{NP} \subseteq \mathbf{QMA}$$

Complexity classes — through concrete problems

We've seen the *definitions*. But what do these classes **feel like**?

Part 2 strategy: understand each class via its complete problem

Class	Complete problem (Part 2)
QMA	k-Local Hamiltonian (ground state energy)
StoqMA	Stoquastic Local Hamiltonian (Ising + transverse field)
BQP	Guided Local Hamiltonian
NP	Commuting Local Hamiltonian

Part 2

Hamiltonian Complexity

The ground-state problem

Physical setup: n quantum spins (qubits) interact locally

$$H = \sum_i h_i, \quad h_i \text{ acts on } \leq k \text{ qubits}$$

The physicist's question: what is the ground-state energy $\lambda_{\min}(H)$?

Examples:

- Heisenberg model: $H = J \sum_{\langle i,j \rangle} (X_i X_j + Y_i Y_j + Z_i Z_j)$
- Quantum Ising: $H = -J \sum_{\langle i,j \rangle} Z_i Z_j - h \sum_i X_i$

Fundamental difficulty: the Hilbert space is 2^n-dimensional and the problem is *quantum*: e.g. unlike classical CSP solutions (n-bit strings), ground states need not have efficient classical descriptions

The k-Local Hamiltonian problem

Definition: given a k-local Hamiltonian $H = \sum_{i=1}^{m} h_i$ where each h_i acts on at most k qubits and $\|h_i\| \leq 1$, and thresholds $a < b$ with $b - a \geq 1/\mathrm{poly}(n)$:

$$\text{Is } \lambda_{\min}(H) \leq a \quad \text{or} \quad \lambda_{\min}(H) \geq b\,?$$

This problem is in QMA:

- **Witness**: the ground state $|\psi_0\rangle$
- **Verification**: measure $\langle h_i \rangle$ for each local term; accept if $\sum_i \langle h_i \rangle \leq a$
    - *(Alternatively: apply QPE on H directly)*
- Quantum computer verifies efficiently — *finding* the ground state may be hard

Kitaev's theorem (2002): k-LH is QMA-*complete* for $k \geq 5$; later $k = 2$ suffices
(Quantum version of Cook-Levin theorem.)

Kitaev's circuit-to-Hamiltonian construction

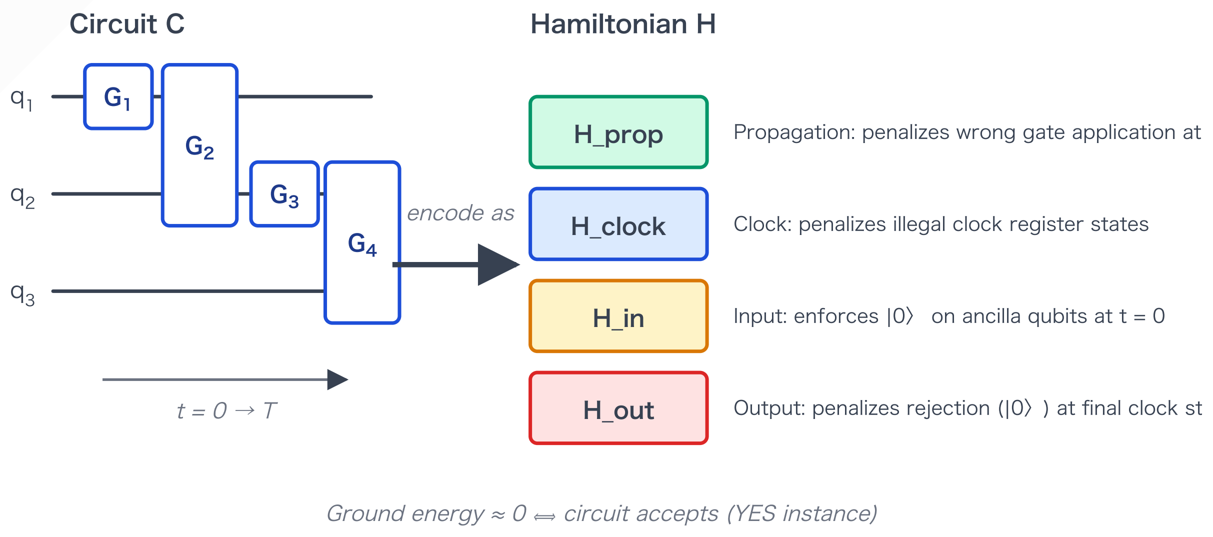
Key idea: encode **correct** quantum computing into its ground state

History state for circuit $C = G_T \cdots G_1$ and witness $|w\rangle$:

$$|\eta(w)\rangle = \frac{1}{\sqrt{T+1}} \sum_{t=0}^{T} |t\rangle_{\text{clock}} \otimes G_t \cdots G_1 |\psi_{\text{in}}(w)\rangle$$

Four Hamiltonian terms:

Term	Role
H_{prop}	Penalizes wrong propagation
H_{clock}	Penalizes illegal clock states
H_{in}	Enforces correct input encoding
H_{out}	Penalizes rejection at final step



The four local terms — explicit definitions

Unary clock: time t encoded as $|1^t 0^{T-t}\rangle_C$ on T clock qubits

$$
\begin{aligned}
H_{\text{clock}} &= \sum_{j=1}^{T-1} |01\rangle\langle 01|_{C_j, C_{j+1}} \\
H_{\text{in}} &= \big(I - |0\cdots 0\rangle\langle 0\cdots 0|\big)_{\text{ancilla}} \otimes |0\rangle\langle 0|_C \\
H_{\text{out}} &= |0\rangle\langle 0|_{\text{out}} \otimes |T\rangle\langle T|_C
\end{aligned}
$$

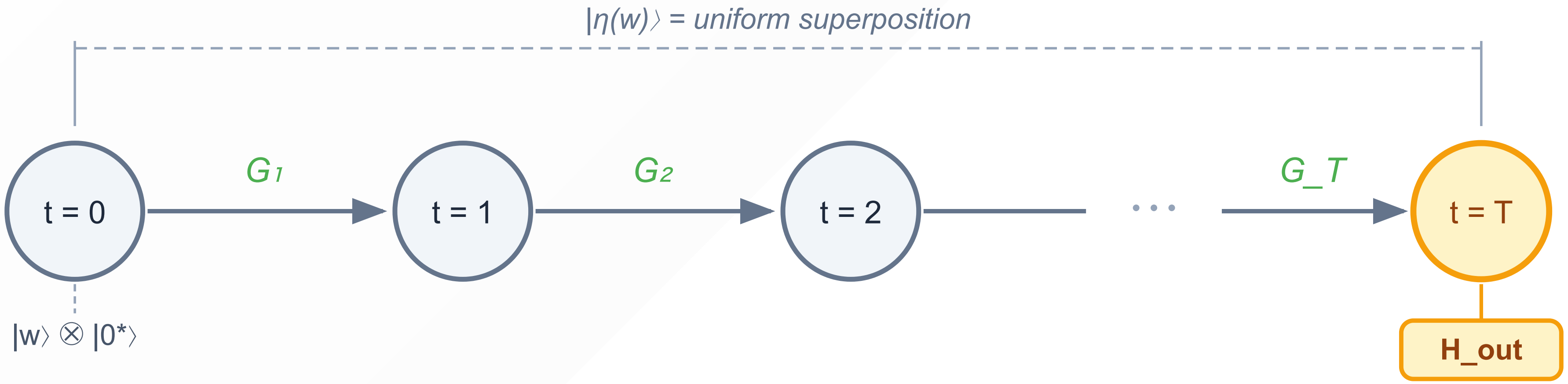
$$
H_{\text{prop}} = \sum_{t=1}^{T} \frac{1}{2}\Big(I \otimes |t\rangle\langle t| + I \otimes |t{-}1\rangle\langle t{-}1| - G_t \otimes |t\rangle\langle t{-}1| - G_t^\dagger \otimes |t{-}1\rangle\langle t|\Big)
$$

All four terms are **positive semidefinite**; $H = H_{\text{clock}} + H_{\text{in}} + H_{\text{out}} + H_{\text{prop}} \geq 0$.

Propagation along a 1D graph of clock states

Zero-energy subspace of $H_{\text{clock}} + H_{\text{in}}$:

$$\ker(H_{\text{clock}} + H_{\text{in}})\big|_{t=0} \;=\; \text{span}\Big\{ \,|0\rangle_{\text{clock}} \otimes |w\rangle \otimes |0^*\rangle \;:\; |w\rangle \in (\mathbb{C}^2)^{\otimes k} \Big\}$$



- H_{prop} propagates along the 1D line graph; each step applies gate G_{t+1}
- $|\eta(w)\rangle$ is the **uniform superposition** over all clock nodes
- H_{out} penalizes the $t{=}T$ node: — checks whether *any* $|w\rangle$ yields acceptance

Completeness and soundness

Completeness (YES instance: $\exists$ witness $|w\rangle$ s.t. circuit accepts with prob. $\geq 2/3$):

$$\exists\,|w\rangle : \quad \langle \eta(w)|H|\eta(w)\rangle = O(1/\mathrm{poly}(n)) \approx 0$$

The history state $|\eta(w)\rangle$ for the good witness is a near-ground state.

Soundness (NO instance: circuit rejects all inputs with prob. $\geq 2/3$):

$$\lambda_{\min}(H) \geq \Omega(1/n^3)$$

Any low-energy state must encode a near-accepting computation — but no such computation exists.

Theorem (Kitaev 2002): The k-Local Hamiltonian problem is QMA-complete for $k \geq 5$.

Theorem (Kempe–Kitaev–Regev 2006): QMA-complete already for $k = 2$.

What QMA-hardness means physically

Computing ground state energy of a generic quantum system is exponentially hard — unless QMA = P (widely believed false)

Consequences:

- No efficient classical algorithm in general
- No efficient quantum algorithm either (unless QMA $\subseteq$ BQP, also unlikely)

But: not all Hamiltonians are equally hard. The complexity landscape is richer.

Part 2b

The Complexity Landscape of Hamiltonians

k-QSAT — quantum satisfiability

k-**QSAT**: given projectors $\Pi_1, \ldots, \Pi_m$ each acting on $\leq k$ qubits, does there exist $|\psi\rangle$ s.t.

$$\Pi_i|\psi\rangle = 0 \quad \forall i \quad (\text{simultaneously satisfiable} = \text{frustration-free})$$

This is k-LH with **perfect completeness**: ground energy exactly 0 vs $\geq 1/\text{poly}(n)$ — i.e., k-LH $\in$ **QMA₁**

k	Complexity	Reference
$k = 2$	**P**	Bravyi 2006
$k = 3$	**QMA₁-complete**	Gosset–Nagaj 2013
$k \geq 4$	**QMA₁-complete**	Bravyi 2006

(for qubits)

QMA₁ vs QMA: whether QMA₁ = QMA is **open** — and gate-set dependent

Stoquastic Hamiltonians and StoqMA

Stoquastic Hamiltonian: all off-diagonal matrix elements in the computational basis are ≤ 0

$$H_{ij} \leq 0 \text{ for all } i \neq j$$

Key property: ground state has all non-negative amplitudes $\langle x|\psi_0\rangle \geq 0$ — **no sign problem**

*(Perron-Frobenius on $G = cI - H \geq 0$; holds for **all** stoquastic H, frustrated or not)*

Examples:

- Transverse-field Ising model: $H = -\sum_{\langle i,j\rangle} Z_i Z_j - h \sum_i X_i$

Complexity:

- StoqMA-complete (Bravyi–Bessen–Terhal (2006))

Stoquastic-k-SAT — MA-completeness

Stoq-k-SAT: given *stoquastic* projectors $\Pi_1, \ldots, \Pi_m$ each on $\leq k$ qubits, is $H = \sum_i \Pi_i$ frustration-free?

Key insight (two independent facts that combine):

1. Stoquastic (Perron-Frobenius) → **any** ground state has non-negative amplitudes — holds generally, not just for FF
2. FF (YES instance) → a zero-energy ground state **exists** as witness

The zero-energy witness has non-negative amplitudes → encodes a **classical probability distribution** → random walk suffices (MA)

k	Complexity	Reference
$k = 2$	**P**	Bravyi 2006
$k \geq 6$	**MA-complete**	Bravyi–Terhal 2008

Commuting Local Hamiltonians

CLH: $[h_i, h_j] = 0$ for all i, j — simultaneous eigenstates exist

Key insight: commutativity allows a *classical witness* — Merlin sends a classical description of which subspace the ground state lives in, and Arthur verifies locally.
(via *Structure Lemma* (Bravyi-Vyalyi 2004))

NP membership — known cases:

Setting	Result	Reference
2-local, any d	**NP** (NP-complete for $d \geq 3$)	Bravyi–Vyalyi 2003
3-local qubits, any graph	in **NP**	Aharonov–Eldar 2011
2D rank-1, any d and k	in **NP**	Bostanci–Hwang 2024
General CLH	**Open** — 20+ years	

Guided Local Hamiltonian

Guided Local Hamiltonian (Gharibian–Le Gall 2022):

- Given: H and a "guiding state" $|c\rangle$ with $|\langle c|\psi_0\rangle|^2 \geq 1/\mathrm{poly}(n)$
- Task: estimate $\lambda_{\min}(H)$
- Complexity: **BQP-complete**

Algorithm: prepare $|c\rangle$, apply amplitude amplification toward the ground state, then QPE

Relevance: models VQE in quantum chemistry — a classical trial state (e.g. Hartree-Fock) as guiding state; a quantum computer can exploit the overlap efficiently

$$\text{Guided LH} \in \mathbf{BQP} \quad \Leftarrow \quad |\langle c|\psi_0\rangle|^2 \geq 1/\mathrm{poly}(n)$$

Improved hardness (Cade–Folkertsma–Gharibian–**Hayakawa**–Le Gall–Morimae–Weggemans, ICALP 2023):

- BQP-complete already for $k = 2$ (optimal locality) with overlap $1 - 1/\mathrm{poly}(n)$
- Holds on 2D square and triangular lattices; also for excited-state energy estimation

Precision changes the complexity

So far, we have considered only $\frac{1}{poly(n)}$ precision;
However, different precision leads to different complexity.

Constant precision (GLH, $\varepsilon = \Omega(1)$) → **classically solvable** (BPP)
QSVT's key subroutine dequantizes at constant precision — Gharibian–Le Gall, STOC 2022

$1/\text{poly}$ **precision** (GLH) → **BQP-complete**
Quantum computer essential — as shown on the previous slide

$1/\exp$ **precision** (k-LH) → **PSPACE-complete**
PreciseQMA = PSPACE — Fefferman–Lin 2016

Spectral density and DQC1

Spectral density of a Hamiltonian H:

$$N_H(a,b) = \frac{1}{2^n}\,\#\{\text{eigenvalues of } H \text{ in } [a,b]\}$$

— fraction of eigenvalues in an interval (not just the minimum).

DQC1-hard (Brandão 2008) — for $O(\log n)$**-local** Hamiltonians:

- **Lower bound**: Feynman–Kitaev construction encodes DQC1 computation into low-energy spectrum of a $\log n$-local H

Why log-local? Kitaev's QMA construction uses a *unary clock* ($|t\rangle$ one-hot encoded), keeping propagation terms 2-local. For DQC1, the unary clock trick is unavailable — a *binary clock* ($\log T$ qubits) is required, making propagation terms $O(\log n)$-local. This is a genuine barrier: DQC1-hardness of spectral density for *constant*-local H is open.

Complexity landscape of Hamiltonians

Class	Complete problem
PSPACE	Precise k-LH ($\delta = 1/\exp$)
QMA	k-Local Hamiltonian (generic)
StoqMA	Stoquastic LH (Ising + transverse field)
QCMA	Guidable LH; Ground state connectivity
BQP	Guided Local Hamiltonian
DQC1	Spectral density of $\log n$-local H
NP	Commuting LH
BPP	1D gapped systems (area law)

Part 2c

Perturbation Gadgets and Universality

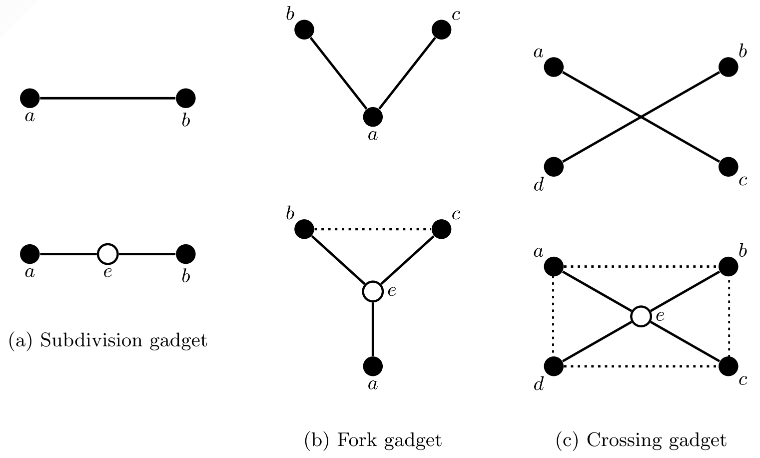
From k-local to 2-local interactions

Idea: encode k-body terms via lower-body interactions with *mediator qubits* + perturbation V, applied iteratively down to 2-local

Setup: $H_{\text{sim}} = \underbrace{\Delta H_0}_{\text{large penalty}} + \underbrace{V}_{\text{2-local}}$

- ΔH_0 confines mediators to a low-energy subspace (energy cost $\geq \Delta$ to excite them)
- Virtual mediator excitations through V generate, at order k in V/Δ, an effective k-body interaction:

$$H_{\text{eff}} \approx H_{\text{target}} + O\left(\frac{\|V\|^{k+1}}{\Delta^k}\right)$$



Three basic gadget types

Physical universality

Universality (Cubitt–Montanaro–Piddock 2018): fixed 2-local interactions can simulate *any* local Hamiltonian

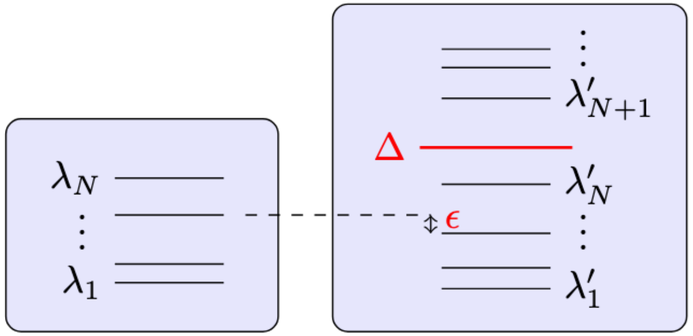


Figure: Cubitt–Montanaro–Piddock (2018)

Complete complexity classification via the **2SLD condition**:
(SLD: Simultaneously Locally Diagonalizable)

$$\text{2SLD: } \exists U \in SU(2) \text{ s.t. } U^{\otimes 2} H_i (U^\dagger)^{\otimes 2} \sim \alpha_i ZZ + A_i \otimes \mathbf{1} + \mathbf{1} \otimes B_i \quad \forall H_i \in S$$

- Non-2SLD → **QMA-complete**; stoquastic-universal → **StoqMA-complete**; 2SLD → **NP-complete**; separable → **P**
- Every 2-local model falls into exactly one class → see next slide

Complexity classification of LH problems

By interaction type (Cubitt–Montanaro–Piddock 2018)

Interaction class	2SLD?	Complexity
Non-2SLD (e.g. Heisenberg $XX+YY+ZZ$, XY $XX+YY$)	No	**QMA-complete**
Stoquastic	Partial	**StoqMA-complete**
2SLD (simultaneously locally diagonalizable)	Yes	**NP-complete**
Fully separable	—	**P**

A quantum analogue of Schaefer's dichotomy — every 2-local model falls into exactly one class.

Remaining directions

- Geometry, topology of interaction
- Complexity with uniform weight
- (Quantum PCP)

Part 2: Key messages

1. The ground-state energy problem is **QMA-complete**

2. The complexity landscape is rich: QMA → StoqMA → BQP → NP → P, each realized by natural Hamiltonians

Part 3

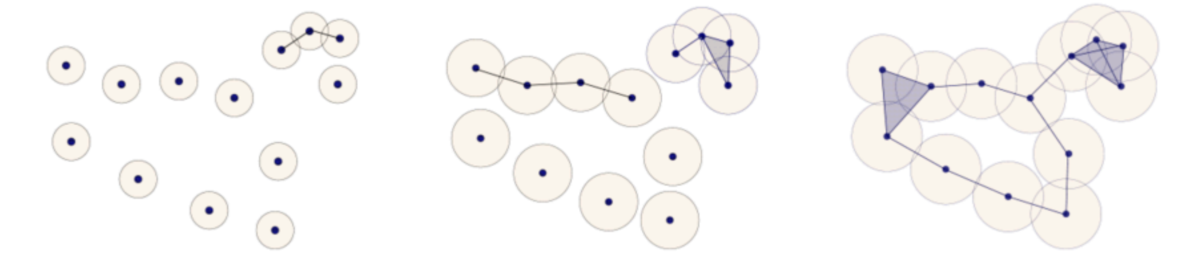
Quantum Topological Data Analysis

From point clouds to topology

Given a **point cloud** in $\mathbb{R}^d$, extract global topological features — robust to noise and small perturbations

Idea: connect nearby points and study the resulting "shape"

- Choose scale ε: connect points within distance ε
- Vary ε from 0 to ∞: topology appears and disappears
- **Persistent features** = signal; **short-lived features** = noise



(a) Rips complex at some scale (b) ..at a larger scale. (c) ...and at an even larger scale.

Unlike PCA (invariant under rotation/scaling), TDA extracts features invariant under bending and stretching

Simplicial complexes and the boundary operator

Simplicial complex K: vertices (0-simplices), edges (1-simplices), triangles (2-simplices), ... closed under taking faces

Clique complex: set of all cliques of graph

Chain group $C_k(K)$: $\mathbb{R}$-linear combinations of oriented k-simplices

Boundary operator $\partial_k : C_k \to C_{k-1}$ — takes the oriented boundary:

$$\partial_1[v_0, v_1] = v_1 - v_0$$

$$\partial_2[v_0, v_1, v_2] = [v_1, v_2] - [v_0, v_2] + [v_0, v_1]$$

Fundamental property: $\partial_k \circ \partial_{k+1} = 0$ — *"the boundary of a boundary is empty"*

$$\cdots \xrightarrow{\partial_{k+2}} C_{k+1} \xrightarrow{\partial_{k+1}} C_k \xrightarrow{\partial_k} C_{k-1} \xrightarrow{\partial_{k-1}} \cdots$$

Homology groups

Cycles $Z_k = \ker \partial_k$: chains with no boundary (closed loops, closed surfaces, ...)

Boundaries $B_k = \mathrm{Im}\, \partial_{k+1}$: chains that bound a higher-dimensional piece

Since $\partial^2 = 0$: $B_k \subseteq Z_k$ — every boundary is a cycle, but not vice versa

k**-th homology group**:

$$H_k(K) \;=\; Z_k \,/\, B_k \;=\; \ker \partial_k \,/\, \mathrm{Im}\, \partial_{k+1}$$

Cycles that cannot be "filled in" — they detect genuine topological holes

Betti number: $\beta_k = \dim H_k(K)$ counts independent k-dimensional holes

Betti numbers — a concrete example

Example: simplicial complex with $\beta_0 = 3, \beta_1 = 1, \beta_2 = 0$

- Three connected components (β_0)
- One independent loops (β_1)
- No enclosed voids (β_2)

Classical cost: computing β_k is #P-hard in general

Can quantum computers help?

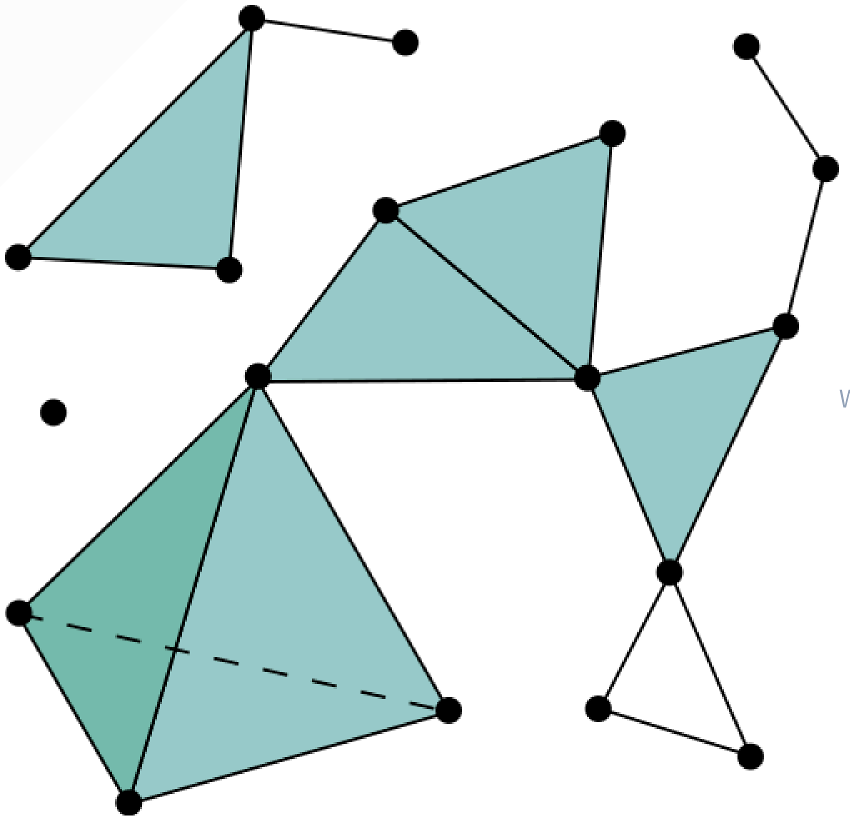


Image: Wikipedia, CC BY-SA 3.0

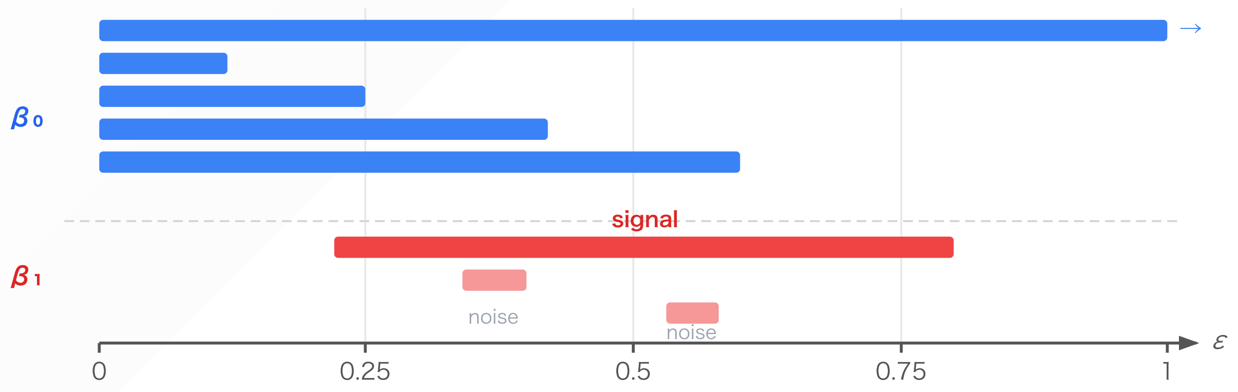
Persistent homology and barcodes

Filtration: a nested sequence of complexes $K_1 \subset K_2 \subset \cdots \subset K_L$ (parameterized by scale ε)

Persistent Betti number $\beta_k^{i,j}$: number of k-dimensional holes present at scale i that survive to scale j

Barcode: each bar $[b, d)$ = a hole born at scale b, dying at scale d

$$\text{long bars} = \text{genuine topology} \qquad \text{short bars} = \text{noise}$$



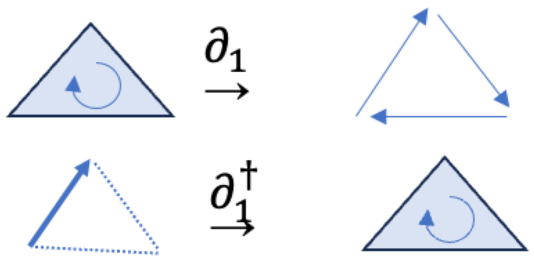
Part 3b

Quantum Algorithms for TDA

The Hodge Laplacian

Combinatorial Laplacian on simplicial complex K:

$$\Delta_k = \partial_k^\dagger \partial_k + \partial_{k+1}\partial_{k+1}^\dagger \;:\; C_k(K) \to C_k(K)$$



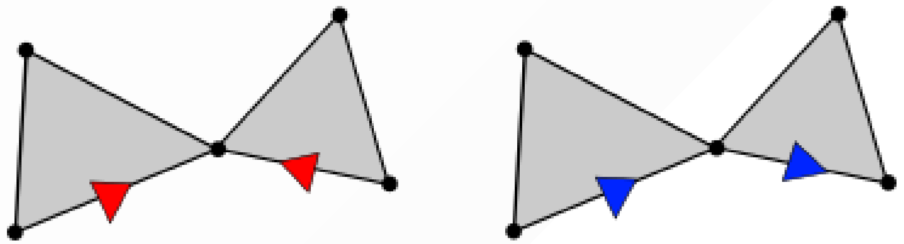
- Symmetric positive semidefinite: $\langle v|\Delta_k|v\rangle \geq 0$ for all v
- Acts on the space $C_k(K)$ of k-chains — an inner product space
- **Analogy**: same structure as a quantum Hamiltonian

Combinatorial Laplacian: matrix form

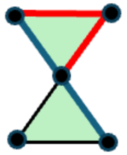
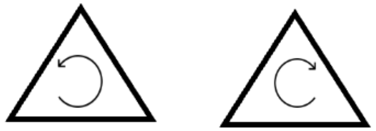
(Goldberg 2002) Indexing d-simplices as $\sigma_1, \ldots, \sigma_m$, the matrix entries of Δ_k are:

$$(\Delta_k)_{ij} = \begin{cases} \deg_U(\sigma_i) + d + 1 & i = j \\ +1 & i \neq j,\ \sigma_i \not\sim^U \sigma_j,\ \text{similar lower adj.} \\ -1 & i \neq j,\ \sigma_i \not\sim^U \sigma_j,\ \text{dissimilar lower adj.} \\ 0 & \text{otherwise (upper adjacent or not lower adjacent)} \end{cases}$$

- $\deg_U(\sigma_i)$: number of $(d+1)$-cofaces of σ_i
- $\sigma_i \sim^U \sigma_j$: upper adjacent (share a common $(d+1)$-coface)
- Similar/dissimilar: the common $(d-1)$-face appears with the same / opposite sign in $\partial_d \sigma_i$ and $\partial_d \sigma_j$



Sign of the combinatorial Laplacian changes with orientation! (Sign problem)

	Graph	Simplicial Complex
Number of objects	$O(n^2)$	$O\binom{n}{k+1}$
Connectivity (adjacency)	Connected by edges (Up adjacency for vertices)	Upper adjacency Connected by $k+1$ -simplices Lower adjacency Connected by $k-1$ -simplices $k=1$ 
Orientation	No inherent orientation on vertices ●	$\pm$ orientation 

→ Homology, unchanged. Changes the sign of Laplacian elements

$\ker \Delta_k \cong H_k(K)$

Since $\Delta_k = \partial_k^\dagger \partial_k + \partial_{k+1}\partial_{k+1}^\dagger$, expanding the inner product:

$$\langle v|\Delta_k|v\rangle = \|\partial_k v\|^2 + \|\partial_{k+1}^\dagger v\|^2$$

Therefore: $\quad \Delta_k v = 0 \iff \partial_k v = 0 \text{ and } \partial_{k+1}^\dagger v = 0$

Condition	Meaning
$\partial_k v = 0$	v is a **cycle** ($v \in Z_k$)
$\partial_{k+1}^\dagger v = 0$	$v \perp \mathrm{Im}\,\partial_{k+1}$ — orthogonal to all boundaries

So $\ker \Delta_k$ = *harmonic forms* = one canonical representative per homology class:

$$C_k = \underbrace{\ker \Delta_k}_{\text{harmonic}} \oplus \underbrace{\mathrm{Im}\,\partial_{k+1}}_{\text{boundary}} \oplus \underbrace{\mathrm{Im}\,\partial_k^\dagger}_{\text{coboundary}} \implies \dim\ker \Delta_k = \beta_k$$

Quantum strategy: Δ_k is PSD like a Hamiltonian → block-encode → QPE → count zero eigenvalues

The LGZ algorithm (Lloyd–Garnerone–Zanardi 2016)

Goal: estimate the normalized Betti number $\beta_k^{\mathrm{norm}} = \beta_k / |X_k|$

Key steps:

1. Prepare mixed state over k-simplices: $\rho_k = \frac{1}{|X_k|} \sum_{\sigma \in X_k} |\sigma\rangle\langle\sigma|$
2. Block-encode Δ_k via QSVT and simulate $e^{-i\Delta_k t}$
3. Apply QPE to sample eigenvalues
4. Estimate frequency of zero eigenvalues $\to \beta_k / |X_k|$

Caveat: efficient only assuming spectral gap $\lambda_{\min}^{>0} = \Omega(1/\mathrm{poly}(n))$

Persistent Betti numbers — Hayakawa (2022)

The LGZ algorithm computes β_k^{norm}, but TDA requires **persistent** Betti numbers:

$$\beta_k^{i,j} = \dim\ker(\partial_k^i) - \dim\Big(\ker(\partial_k^i) \cap \mathrm{Im}(\partial_{k+1}^j)\Big)$$

This counts holes born at scale i that survive to scale j — the key quantity for barcodes.

Algorithm (Hayakawa 2022):

- Block-encode the **persistent Hodge Laplacian** $\Delta_k^{i,j}$, whose kernel has dimension $\beta_k^{i,j}$

$$\Delta_k^{i,j} := \partial_{k+1}^{j,i} \circ \left(\partial_{k+1}^{j,i}\right)^* + \left(\partial_k^i\right)^* \circ \partial_k^i$$

- Apply QPE to estimate fraction of zero eigenvalues

Result: polynomial-time quantum algorithm for $\beta_k^{i,j}$ (normalized), enabling full barcode computation

Part 3c

The Complexity Landscape of QTDA

Quantumness in computational topology

Classical hardness of Betti numbers: a long-standing open problem

Theorem (Adamaszek–Stacho 2017): computing homology of clique complexes is **NP-hard**
— settled after decades of effort

Then came a quantum surprise:

Theorem (Crichigno–Kohler 2022): Clique Homology (gapped) is **QMA_1-complete**
— not just classically hard, but *quantum* hard

This opened a new question:

What is the full quantum complexity landscape of TDA problems?
Can we find problems at every level — BQP, MA, QMA, ...?

This section: map out where TDA problems sit across the quantum hierarchy
Also: does QTDA (e.g., LGZ) actually offer *quantum advantage* over classical algorithms?

Hardness proof strategy: Hamiltonian → Homology

Key idea (used in all QMA/MA hardness reductions):

Step 1 — Encode the Hilbert space

Build a simplicial complex K_0 whose harmonic subspace $\ker \Delta_k(K_0)$ has dimension 2^n:

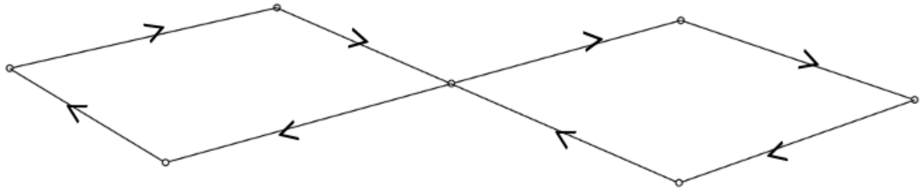
$$\ker \Delta_k(K_0) \;\cong\; (\mathbb{C}^2)^{\otimes n} \quad (n\text{-qubit Hilbert space})$$

Step 2 — Encode the Hamiltonian

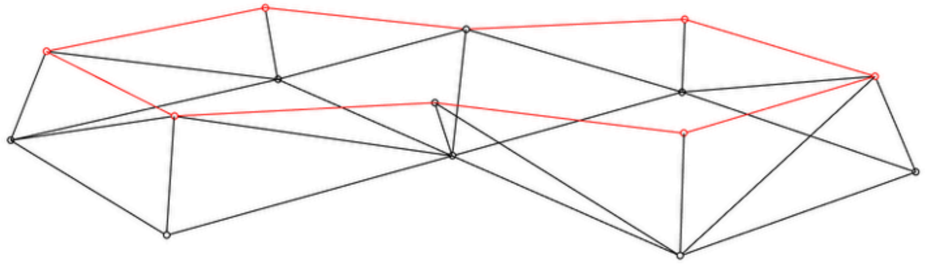
For each local term h_i in $H = \sum_i h_i$, attach simplicial gadgets that kill harmonic cycles in directions penalized by h_i:

$$\beta_k(K) > 0 \quad \Longleftrightarrow \quad H \text{ has a zero-energy ground state}$$

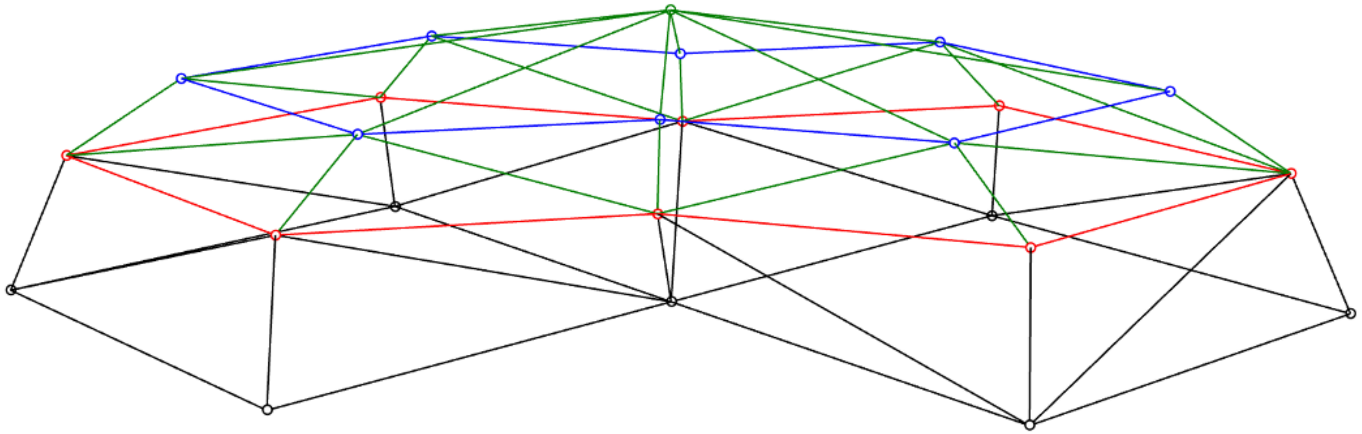
Example: fill a hole correspond to $|0\rangle + |1\rangle$



$|0\rangle + |1\rangle$

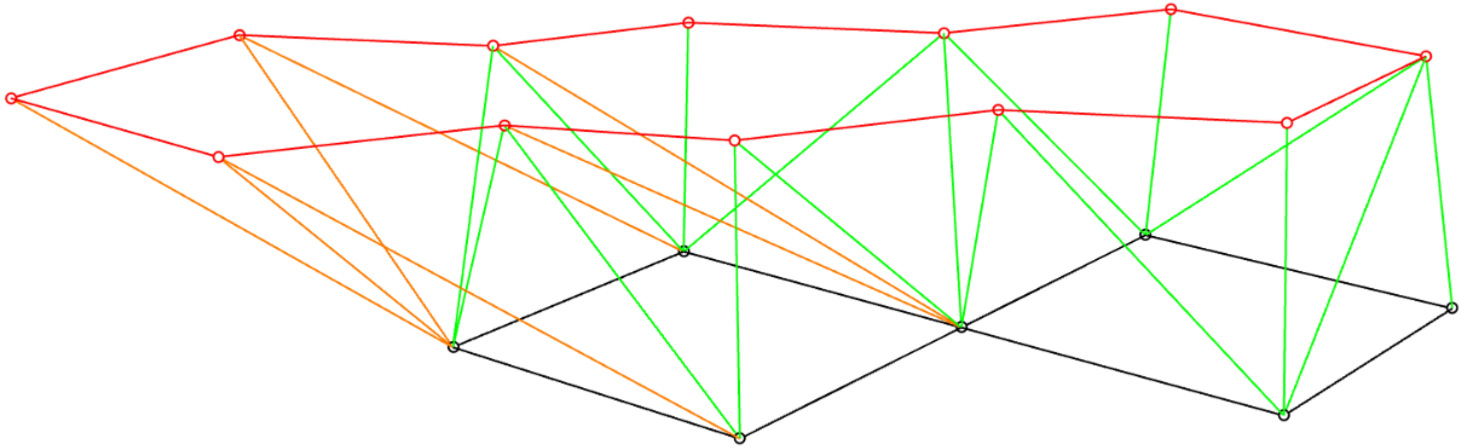


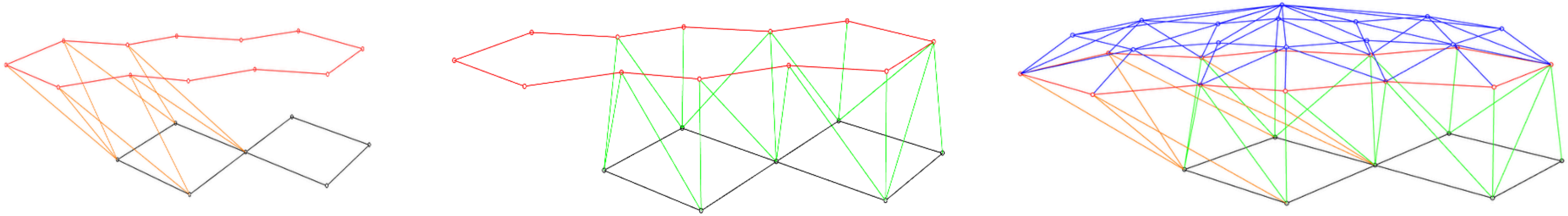
Copy of $|0\rangle + |1\rangle$



Make a triangulated roof

Example: integer hole $2|0\rangle + |1\rangle$





Harmonic Persistence is BQP-complete

Gyurik, Hayakawa, Schmidhuber, King, Dunjko (2024)

Harmonic Persistence problem:

Given $K_1 \subset K_2$, does there exist a harmonic k-cycle of K_1 that remains harmonic in K_2?

Theorem: Harmonic Persistence is **BQP-complete** (BQP$_1$-hard and in BQP)

Proof structure:

- **Contained in BQP**: efficient quantum simulation of Hodge Laplacian exists
- **BQP$_1$-hard**: reduction from a BQP-complete problem with one-sided error

Meaning: if a classical computer could solve it → $\mathsf{BQP} = \mathsf{P}$. A problem *about topology* that is *intrinsically* quantum-computational.

MA-completeness with Orientable Filtration

Hayakawa, Gyurik, Rad, Dunjko (2025)

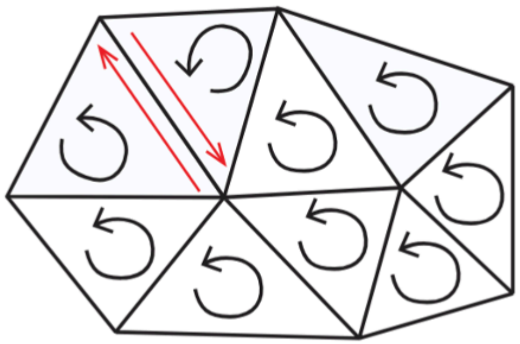
Problem: given a uniform orientable filtration, decide whether a homology class is trivial

Theorem: This problem is **MA-complete**

- **Containment** (MA): random walks on simplicial complexes via sign-problem-free Hamiltonian (from orientation data)
- **Hardness** (MA-hard): reduction from stoquastic Local Hamiltonian (StoqMA-complete)

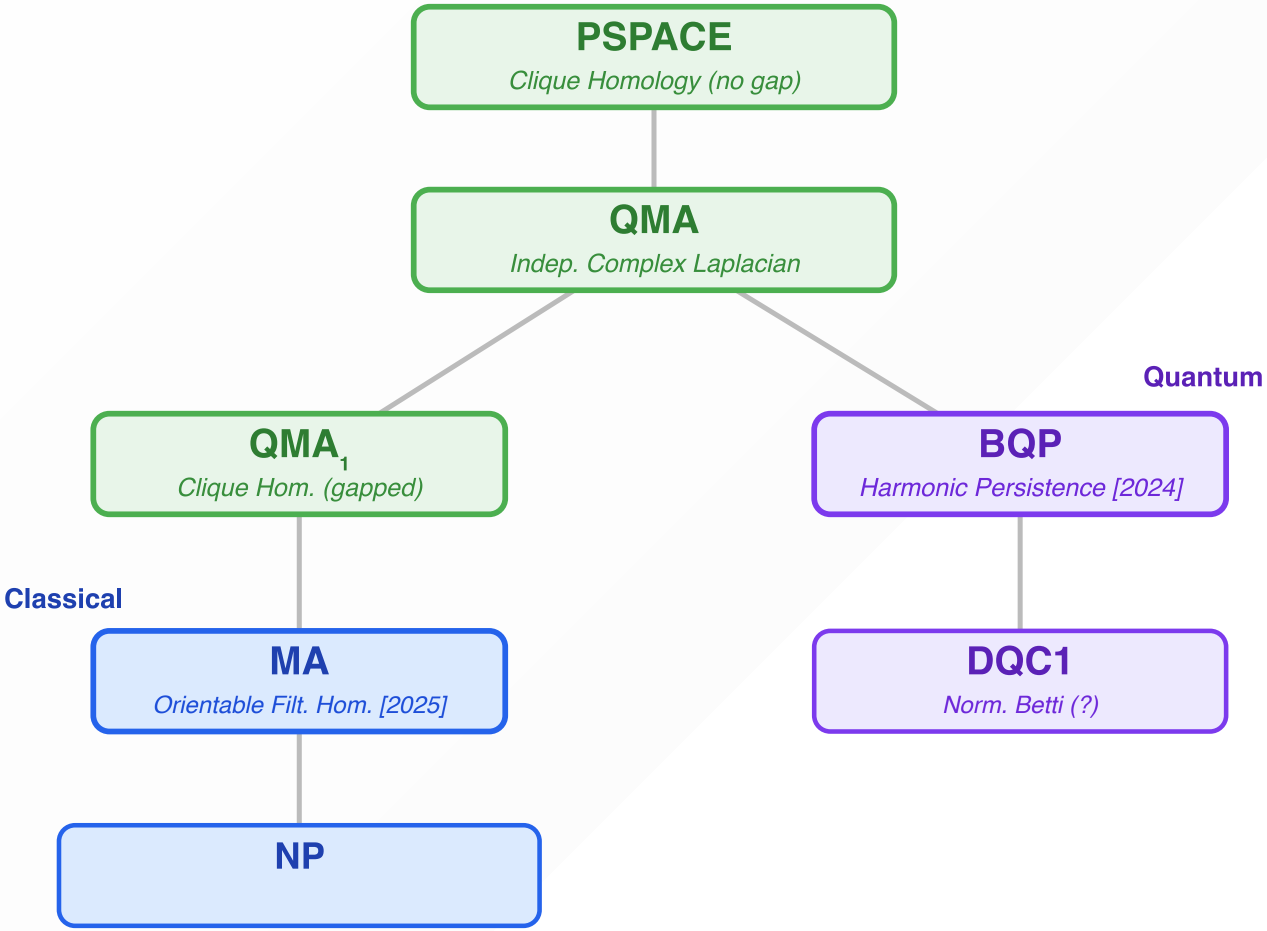
Significance: natural TDA problem where classical *randomness* (but not determinism) suffices — distinct from BQP

d-Orientable



Discretization of manifold

The full complexity landscape of QTDA



- **PSPACE**: Clique Homology (no gap) — Rudolph 2025
- **QMA**: Independent Complex Laplacian — Rayudu 2025
- **QMA$_1$**: Clique Homology (gapped) — Crichigno–Kohler 2022
- **MA**: Orientable Filtration Homology — **Hayakawa et al. 2025**
- **BQP**: Harmonic Persistence — **Gyurik–Hayakawa et al. 2024**
- **DQC1**: Normalized Betti numbers — Cade 2021, Gyurik 2022
- **#P / NP**: Exact / multiplicative Betti numbers

QTDA: complexity landscape and open problems

The full landscape (as of 2025)

Problem	Class	Reference
Clique Homology (no gap)	**PSPACE**	Rudolph 2025
Independent complex Laplacian spectral gap	**QMA**	Rayudu 2025
Clique Homology (gapped)	**QMA_1**	Crichigno–Kohler 2022
Orientable Filtration Homology	**MA**	Hayakawa et al. 2025
Harmonic Persistence	**BQP**	Gyurik–Hayakawa et al. 2024
Normalized Betti numbers (algorithm)	**DQC1**	Cade 2021; Gyurik 2022
Exact Betti numbers	**#P-hard**	Lloyd–Schmidhuber 2023

Central open problem: is normalized Betti number estimation **DQC1-hard**?

Interlude

Topology Beyond TDA: Berry Phase

Berry phase — a topological invariant in quantum matter

For a Hamiltonian $H(\lambda)$ with $\lambda \in [0,1]$, $H(0) = H(1)$, the non-degenerate ground state $|\psi(\lambda)\rangle$ acquires a phase after adiabatic evolution around the loop:

$$U(T)|\psi(0)\rangle = e^{-i\theta_D}\, e^{i\theta_B}\, |\psi(0)\rangle$$

- **Dynamical phase** θ_D: depends on speed and energy — removable
- **Berry phase** $\theta_B = i \oint \langle \psi | \nabla_\lambda | \psi \rangle \cdot d\lambda$: depends only on the **geometry of the loop**

θ_B is a fundamental topological invariant: $\mathbb{Z}_2$ invariant of the SSH model, foundation of the Chern number, bulk-boundary correspondence, ...

But what is the computational complexity of estimating the Berry phase for strongly correlated systems?

Complexity of Berry phase estimation

Hayakawa, Sakamoto & Kiumi (2025)

Setting	Extra input	Complexity
Guiding state known	$\vert c\rangle$ with $\vert\langle c\vert\psi(0)\rangle\vert^2 \geq 1/\mathrm{poly}(n)$	**BQP-complete**
Energy threshold known	$E_0 < E_{\mathrm{th}} < E_1$	**dUQMA-complete**
No additional input	—	$\mathsf{P}^{\mathsf{dUQMA}[\log]}$-hard

- **BQP-completeness**: Berry phase estimation in topological matter is *intrinsically quantum* — no classical polynomial-time algorithm unless BQP = P
- **dUQMA** (doubly-Unique QMA): new complexity class introduced here — $\mathsf{BQP} \subseteq \mathsf{dUQMA} = \text{co-}\mathsf{dUQMA} \subseteq \mathsf{QMA}$

New quantum algorithm: eliminates the dynamical phase by running two adiabatic evolutions at different energy scales, then taking the difference

Quantum algorithm for Berry phase estimationx

Step 1: phase estimation on adiabatic evolution $U(T)$ for time T:

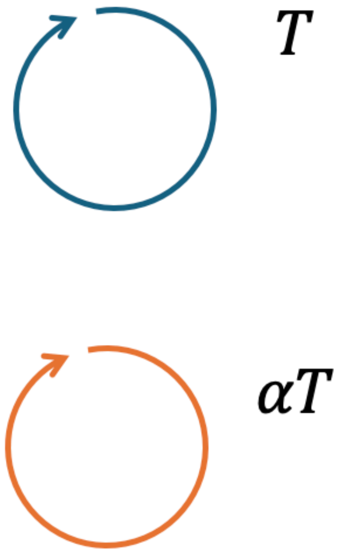
$$\underline{\varphi_1 = \theta_B + \theta_D} \pmod{2\pi}.$$

Step 2: repeat the process with a different runtime αT:

$$\underline{\varphi_\alpha = \theta_B + \alpha\theta_D} \pmod{2\pi}.$$

Step 3: α must be chosen so that $\varphi_\alpha - \varphi_1 = (\alpha - 1)\theta_D \in [-\pi, \pi)$, allowing a **unique lift** from mod 2π to a real number. Then,

$$\theta_B = \varphi_1 - \theta_D \pmod{2\pi}.$$



→ Classically post-process to estimate $\boldsymbol{\theta_B}$

Topology in physics: two complementary directions

	Quantum TDA	Berry phase complexity
Topology type	Data topology (barcodes, Betti numbers)	Quantum state topology (Berry phase, Chern number)
Input	Classical point cloud	Local Hamiltonian
Task	Count topological holes in data	Estimate geometric phase
Status	Exponential speedup exists (worst-case)	Exponential speedup exists (worst-case)

Conclusion

What we've seen today

Part 1: Quantum computation basics

- QPE, amplitude amplification, QSVT: the three tools that unify quantum algorithms
- P $\subseteq$ BQP $\subseteq$ QMA — each class realized by natural problems

Part 2: Hamiltonian complexity

- The k-Local Hamiltonian problem is **QMA-complete** (Kitaev 2002, Kempe–Kitaev–Regev 2006)
- The complexity landscape: QMA $\to$ StoqMA $\to$ BQP $\to$ NP $\to$ P

Part 3: Quantum TDA

- Topology $\to$ Hodge Laplacian $\to$ QPE: the LGZ algorithm and its extensions
- QTDA problems span **DQC1 to PSPACE** — a richer landscape than expected
- Complexity hardness results (BQP, MA, QMA) confirm quantum hardness; quantum advantage for practical tasks remains an open question

Thank you

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